A3-PS033-TC395

Secure DNS-Based Discovery of Diagnostic Services Using DNSSEC

Team Name: Team Phoenix

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Problem Description

When something goes wrong on the network, we need to know where the diagnostic tools are — and whether we can trust what DNS tells us about them.

- Latency, traceroute and health services are useful only if a client can **find** them. Typing addresses in by hand is slow and breaks when services move.
- DNS answers can be altered on the way. A client that accepts a changed answer could be sent to the wrong endpoint without noticing.
- **Our question:** can DNS TXT records carry the service details, and can DNSSEC signatures let the client check that data before using it?
- We built a small local prototype to test exactly this, rather than a full production system.

Background

The three pieces our prototype relies on.

- **DNS TXT records** hold short pieces of text under a domain name, so they can store structured service details such as an endpoint path and its type [1][2].
- **DNSSEC** adds digital signatures to DNS data. A signed answer comes with an RRSIG record, and the matching public key is published as a DNSKEY record [3][4].
- **Validation** means recomputing the signature check on the answer. If the data was changed, the check fails and the client should refuse it.
- A complete DNSSEC check also follows the **chain of trust** from the root down to the zone. Our prototype checks signatures with one configured key and does not yet do this full chain.

Literature & RFC Review

We looked at the standards most relevant to publishing and trusting service information in DNS.

- **RFC 1035** – defines the DNS message format and the TXT record type we use for metadata [1].
- **RFC 1464** – describes storing arbitrary string attributes in DNS TXT records [2].
- **RFC 4033 / 4034 / 4035** – the DNSSEC specifications, including RRSIG and DNSKEY records and how validation works [3][4][5].
- **RFC 6763** – DNS-Based Service Discovery, the established approach for finding services through DNS [6].
- **Gap we noticed:** these standards cover discovery and signing separately. Our project tries them together for network diagnostic endpoints.

Proposed Solution

Store the service details in DNS, sign them, and let the client verify before it trusts anything.

- **DNS zone:** TXT records describe the latency, traceroute and health services.
- **DNS test server:** answers queries from the zone. In our tests it ran locally at `127.0.0.1:5353`.
- **Discovery client:** asks for a service, reads the JSON metadata, and checks the RRSIG with the configured DNSKEY.
- **Diagnostic services:** the endpoints `/latency`, `/traceroute` and `/health` that the metadata points to.
- If the signature check fails, the client rejects the record instead of using it.

Optimization Proposed

What changes compared with entering and trusting service addresses by hand.

Aspect	Before	With our approach
Finding services	Addresses typed in manually	Client looks them up in DNS TXT records
Managing changes	Update every client by hand	Update the DNS record in one place
Trusting the data	Answer accepted as received	Signature checked before use
Altered metadata	May go unnoticed	Rejected in our tampering test
Trust chain	Not applicable	Not yet verified (planned next step)

Solution Architecture & Design

Four components, one request path.

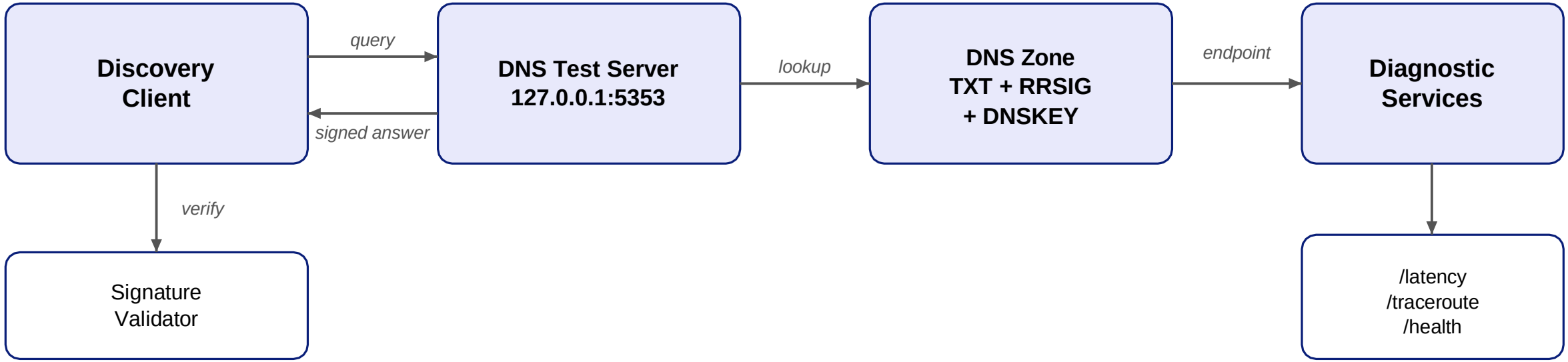


Figure: System architecture of the prototype

Data Flow Diagram

How a service is discovered and checked, step by step.

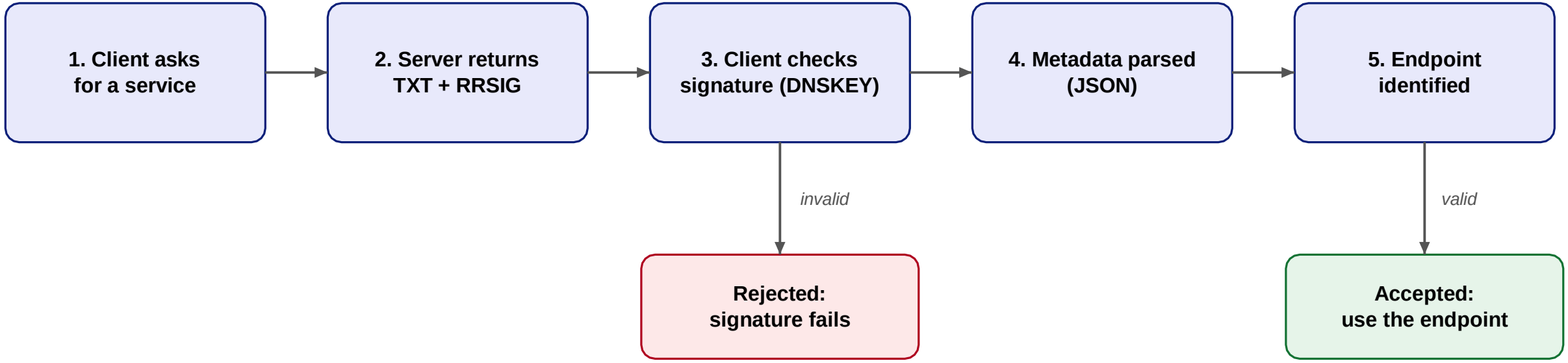


Figure: Data flow of the discovery and validation process

Deployment Timeline

Where the project stands and the order we would take next.

Done

Local prototype

TXT-based discovery of three services, signature check with a configured DNSKEY, 14 passing tests.

Next

Trust chain

Add complete DNSSEC chain-of-trust validation.

Then

Real DNS server

Set up a properly configured authoritative DNS server.

Later

Real tools & keys

Connect real diagnostic tools; add key management and signature-expiry monitoring; test more failure cases.

Bibliography

1 P. Mockapetris. *Domain Names – Implementation and Specification*. RFC 1035, 1987.

2 R. Rosenbaum. *Using the Domain Name System To Store Arbitrary String Attributes*. RFC 1464, 1993.

3 R. Arends et al. *DNS Security Introduction and Requirements*. RFC 4033, 2005.

4 R. Arends et al. *Resource Records for the DNS Security Extensions*. RFC 4034, 2005.

5 R. Arends et al. *Protocol Modifications for the DNS Security Extensions*. RFC 4035, 2005.

6 S. Cheshire, M. Krochmal. *DNS-Based Service Discovery*. RFC 6763, 2013.

Thank You for Your Attention!